

MATERIALS SCIENCE

Radiation-resilient monolithic wide-bandgap perovskite/p-type heterojunction silicon tandem solar cells for space photovoltaics

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Perovskite/silicon tandem solar cells (TSCs) are attractive candidates for efficient, low-cost space photovoltaics, but radiation tolerance and operational stability limit deployment. Here, a monolithic tandem architecture integrating a stabilized 1.72-electron volt wide-bandgap perovskite top cell with a radiation-hardened p-type heterojunction silicon bottom cell is reported. The perovskite absorber is stabilized by a multifunctional ionic liquid additive, which enhances crystallinity, suppresses phase segregation, and improves thermal and photostability, resulting in a power conversion efficiency of 24.0% and a certified value of 23.58%. The resulting TSCs exhibit a certified zero air mass efficiency of 27.49% (12.56 square centimeters). Under 1 mega-electron volt electron irradiation at 1×10^{14} electrons per square centimeter, the TSC retains nearly 80% of its initial performance, whereas under 150 kilo-electron volt proton irradiation at 1×10^{12} protons per square centimeter, it retains 93% of its initial performance, accompanied by a recoverable response. A high-altitude balloon campaign further records a stable power output of up to 387.4 milliwatt at ~30-kilometer altitude. This work demonstrates a viable pathway for next-generation space photovoltaics.

INTRODUCTION

The rapid deployment of satellite constellations has intensified the demand for photovoltaic technologies that offer high efficiency, low specific mass, and long-term stability under harsh space conditions. This trend is particularly evident in the emergence of large-scale low Earth orbit (LEO) constellations, which are fundamentally reshaping the space industry by imposing stringent constraints on cost, scalability, and performance. Although III-V multijunction solar cells remain the benchmark for space photovoltaics because of their high efficiency, outstanding radiation hardness, and proven flight heritage, their widespread deployment is constrained by complex fabrication and high manufacturing cost (1–3). At the other end, conventional silicon solar cells offer lower cost and mature manufacturing, but their specific power and radiation tolerance are comparatively limited for demanding space applications. Recently, emerging thin-film solar cells such as organic-inorganic halide perovskites (4–10), organic photovoltaic devices (11–13), and Cu(In, Ga)Se₂ thin-film solar cells (14–16) have attracted attention for space applications due to their low-cost, lightweight structure, mechanical flexibility, and excellent scalability. Notably, lead halide perovskite solar cells (PSCs)

have demonstrated substantial resistance to electron and proton bombardment in LEO-simulated environments (17–23). However, most single-junction devices still suffer from relatively low power conversion efficiencies (PCEs), typically below 20% under AM0 (zero air mass) conditions, which limits their deployment in high power-demanding missions.

Perovskite/silicon TSCs address this limitation by combining the radiation tolerance of perovskites with the mature device platform and near-infrared response of silicon, enabling efficient power generation in space, with AM0 efficiencies of up to 21.1% reported (24). Despite the compelling potential of perovskite/silicon TSCs to achieve high PCEs under AM0 conditions, less attention has been paid to their radiation resilience and the severe thermal fluctuations they face. These aspects are crucial for ensuring long-term functionality in LEO orbit, where devices are subject to harsh conditions (figs. S1 and S2), including thermal cycling and high-energy particle bombardment, which causes semiconductor damage through ionization and atomic displacement (25, 26). In monolithic TSCs, the series-connected architecture makes the overall device performance directly constrained by radiation-induced degradation in either subcell. This issue is particularly critical for the Si bottom cell, whose radiation tolerance depends strongly on both conductivity type and thickness. Compared with conventional n-type Si (n-Si), p-type Si (p-Si) generally exhibits superior space-radiation hardness (27, 28), and this tolerance can be further enhanced by reducing the absorber thickness (29, 30). These considerations underscore the urgent need for tandem design strategies that simultaneously deliver high AM0 efficiency and robust radiation tolerance at the device level.

In this work, we develop a radiation-resilient monolithic tandem architecture that integrates a compositionally optimized wide-bandgap (WBG) 1.72 eV formamidinium (FA)/Cs-based perovskite top cell with a 100- μ m-thick p-Si bottom cell. Notably, this design not only improves spectral matching under AM0 illumination, but

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also leverages the thin p-Si subcell to enhance space relevance. To stabilize the WBG perovskite, we incorporate the ionic liquid additive 1,3-bis(cyanomethyl)imidazolium chloride ([Bcmim]Cl), which improves crystallization and reduces trap density. As a result, the perovskite films show improved crystallinity, reduced trap density, and enhanced resistance to thermal, optical, and irradiation-induced degradation. In particular, under 150-keV proton irradiation, the additive-treated films with lower defect exhibit improved retention of the initial efficiency at 1×10^{12} and 1×10^{13} p^+/cm^2 fluence. Last, the optimized tandem architecture achieves efficiencies of 32.07% (0.95 cm^2) and 30.58% (12.56 cm^2) under AM1.5G (Air Mass 1.5 Global), along with a certified efficiency of 27.49% under AM0 illumination, representing the highest certified AM0 efficiency reported to date for perovskite/silicon TSCs. Under 1-MeV electron irradiation ($1 \times 10^{14} \text{ e}^-/\text{cm}^2$), the optimized TSCs retain nearly 80% of their initial efficiency, while under 150-keV proton irradiation, the TSC retains 93% of its initial efficiency at $1 \times 10^{12} \text{ p}^+/\text{cm}^2$. These results demonstrate a notable improvement in radiation stability compared to state-of-the-art n-Si-based TSCs. Beyond ground-based qualification, we further demonstrated near-space in situ operation via a high-altitude balloon campaign, capturing real-time electrical outputs and stable power delivery up to 387.4 mW at 20- to 30-km altitude.

RESULTS

Electron irradiation effects in perovskite/silicon tandems

To investigate radiation damage, we used 1-MeV electrons as a benchmark to assess the radiation response of the monolithic tandem, given their widespread use in space-solar-cell radiation studies, thereby enabling direct comparison with the established literature on Si and III-V photovoltaics. To elucidate the interaction depth and energy-deposition profile, we performed Monte Carlo simulations (CASINO) of 1-MeV electrons incident on the perovskite/heterojunction (HJT) silicon monolithic tandem architecture. Basic information for each functional layer in the simulated device is listed in table S1. As shown in Fig. 1A and fig. S3, the simulated stack mainly comprised an $\sim 1\text{-}\mu\text{m}$ perovskite top cell and an $\sim 100\text{-}\mu\text{m}$ silicon bottom cell. The simulations reveal that a modest fraction of the incident electron energy is deposited within the shallow perovskite layer, whereas the silicon subcell absorbs the majority. Owing to their low mass and high penetration capability, energetic electrons traverse the entire device, producing a broad energy-deposition profile with a maximum at $\sim 60 \mu\text{m}$ beneath the silicon surface. For clarity, each constituent layer, including the perovskite and p-Si absorber, is indicated by arrows and labeled to correlate the deposition profile with the device architecture directly.

To assess the role of the bottom cell, we further examined the radiation tolerance of Si solar cells. We conducted 1-MeV electron

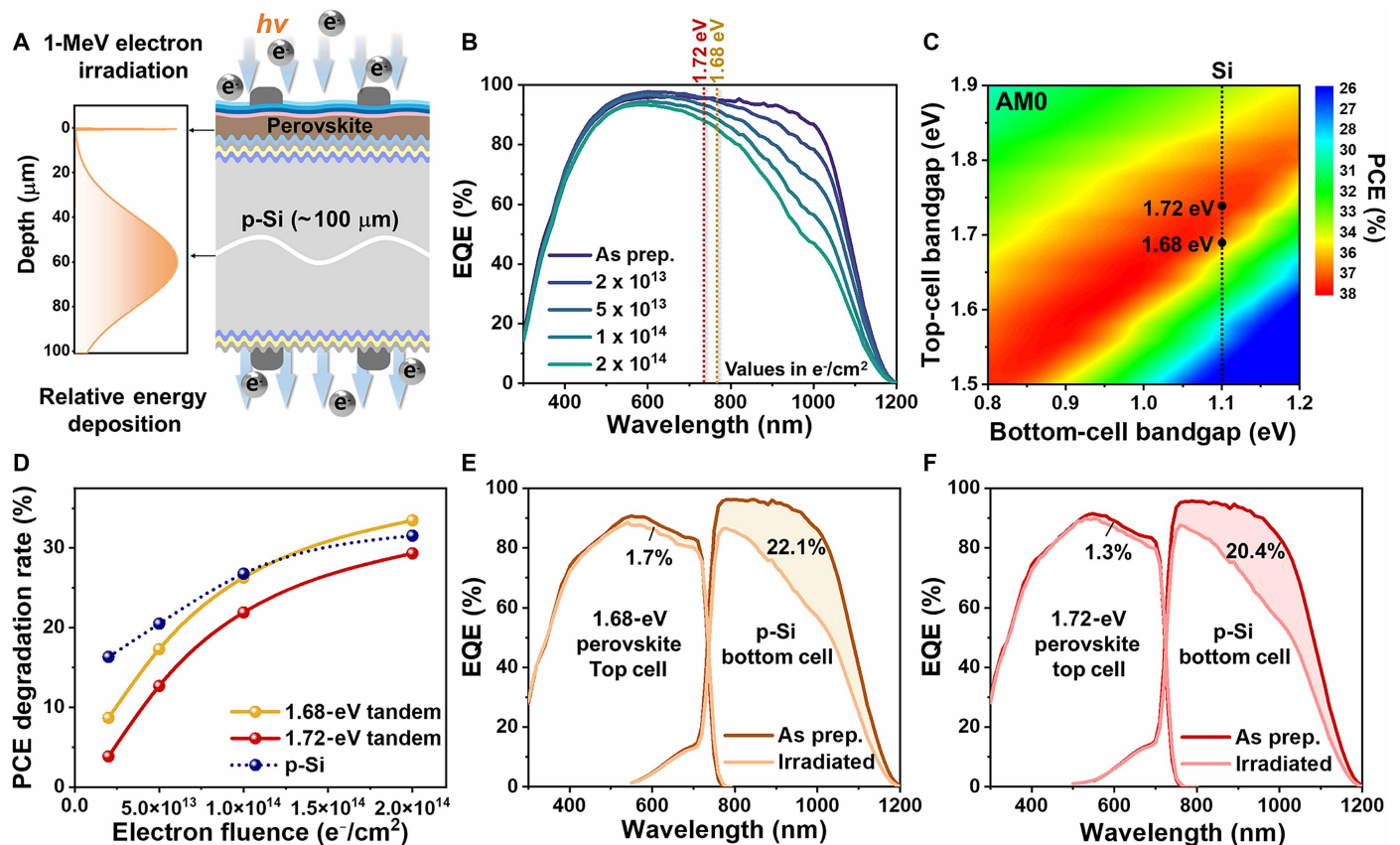


Fig. 1. Electron radiation tolerance of monolithic perovskite/Si TSCs with varying top-cell bandgaps. (A) Simulated energy deposition profile of 1-MeV electrons in the perovskite/silicon TSC, along with the corresponding schematic of the device structure. (B) EQE spectra of p-Si solar cells before and after 1-MeV electron irradiation at different fluences. (C) Simulated maximum theoretical PCE contour maps for two-terminal perovskite/silicon tandems as a function of top-cell bandgap and bottom-cell current density under AM0 spectrum. (D) Efficiency degradation rate of different devices after exposure to 1-MeV electrons. (E and F) EQE spectra of monolithic TSCs with 1.68-eV and 1.72-eV perovskite top cells before and after $1 \times 10^{14} \text{ e}^-/\text{cm}^2$ electron irradiation.

irradiation experiments on both p-type and n-type HJT silicon solar cells at fluences of 2×10^{13} , 5×10^{13} , 1×10^{14} , and 2×10^{14} e⁻/cm². All electron irradiation experiments in this study were performed on unencapsulated devices. The n-type HJT silicon structure used in this study corresponds to that of commercially available devices widely adopted in state-of-the-art tandem architectures (31–35). Based on the results shown in fig. S4, the n-Si devices suffered a near-complete loss of performance after 1-MeV electron irradiation. The short-circuit current density (J_{SC}) dropped dramatically from 47.08 to 1.96 mA/cm² at a fluence of 2×10^{14} e⁻/cm², accompanied by severe degradation in open-circuit voltage (V_{OC}) and overall efficiency. The efficiency degradation rate reached 97.58% at the highest fluence, as summarized in fig. S5, underscoring the poor radiation tolerance of conventional n-Si devices under space-relevant electron exposure. In contrast, the p-Si solar cells demonstrated improved radiation tolerance, with PCE losses of 16.8, 21.3, 27.4, and 31.8% at the respective fluences (fig. S6). These results reveal a clear fluence-dependent degradation trend, which likely originates from cumulative radiation-induced defect formation that limits both carrier generation and recombination dynamics. To further elucidate this behavior, we analyzed external quantum efficiency (EQE) spectra under varying irradiation fluences (Fig. 1B). A pronounced, fluence-dependent suppression is observed in the long-wavelength region above 700 nm. Spectral dependence of the relative change of EQE as a function of electron fluence is shown in fig. S7. At $\lambda = 700$ nm, the optical absorption coefficient of crystalline silicon is $\sim 1.5 \times 10^3$ cm⁻¹, corresponding to an absorption depth of ~ 5 to 7 μ m (36). This indicates that long-wavelength photons are primarily absorbed beyond the depletion region, within the quasi-neutral region of the silicon bulk. The observed long-wavelength EQE loss therefore indicates enhanced defect-assisted bulk recombination and reduced minority-carrier diffusion length in irradiated silicon.

Therefore, to mitigate the enhanced long-wavelength degradation of irradiated silicon, we systematically engineered the perovskite top-cell bandgap to preserve current matching and tandem performance under space-relevant radiation conditions. Under AM0 illumination, where photon flux between 300 and 700 nm is elevated (fig. S8), our optical simulations (Fig. 1C) identify 1.72 eV as the optimal top-cell bandgap, yielding superior current balance with silicon and enabling AM0 efficiencies approaching 38%. To substantiate the bandgap selection, we compared the absorption onsets of 1.68- and 1.72-eV perovskite films with the EQE spectrum of the silicon bottom cell (Fig. 1B). Although 1.68-eV perovskites are commonly used in tandem architectures (31, 35) for their proven efficiency under AM1.5G, the silicon subcell exhibits fluence-dependent EQE degradation that is more pronounced at longer wavelengths. Widening the top-cell bandgap to 1.72 eV increases the fraction of near-infrared photons transmitted to the silicon layer, mitigating radiation-induced photocurrent losses and improving current matching under space-relevant conditions. To validate this design, we first fabricated single-junction WBG PSCs with different bandgaps (1.68 and 1.72 eV) and quantitatively confirmed these values by analyzing the first derivative of each EQE spectrum (fig. S9). We then fabricated monolithic tandem devices incorporating these WBG PSCs and subjected them to 1-MeV electron irradiation at varying fluences. After irradiation, both TSCs became silicon-limited, but mismatch losses were considerably reduced in the 1.72-eV device. The 1.72-eV tandem retained nearly 80.0% of its initial PCE, while the 1.68-eV counterpart suffered 27.0% efficiency loss at a fluence of 1×10^{14} e⁻/cm² (Fig. 1D). Quantitative

analysis of integrated subcell photocurrents derived from the EQE data (Fig. 1, E and F) before and after 1×10^{14} e⁻/cm² electron exposure shows only minor current loss in the perovskite absorbers: 1.7% for the 1.68-eV tandem and 1.3% for the 1.72-eV tandem. The corresponding silicon subcell currents decreased by 22.1 and 20.4%, respectively. These results show that the irradiation-induced performance loss is mainly governed by the silicon bottom cell. The lower top-cell photocurrent in the 1.72-eV tandem leads to more favorable current matching after silicon degradation, thereby improving the retained tandem photocurrent after irradiation.

Engineering of WBG perovskite films

Motivated by the improved current matching and radiation performance achieved with a 1.72-eV top cell, we implemented this WBG perovskite in our experimental devices. However, we noted that WBG perovskites (>1.70 eV) commonly experience stability issues, including phase segregation and increased nonradiative losses, which are further exacerbated in harsh space environments (37–40). To address these challenges, we incorporated a functional Lewis-basic ionic-liquid additive, [Bcmim]Cl, into the perovskite precursor formulation. The average grain size increased after the additive treatment as observed by scanning electron microscopy (SEM) (Fig. 2A and figs. S10 and S11), accompanied by a narrower size distribution, indicating not only enhanced crystallinity but also improved grain size uniformity. We also performed the grazing-incidence wide-angle x-ray scattering (GIWAXS) measurements on films with and without the additive (Fig. 2B). Upon incorporating [Bcmim]Cl, the PbI₂-related signal ($0.9 \text{ \AA}^{-1}$) observed in the control device was substantially suppressed, while the intensities of the (100) and (200) crystallographic planes increased, indicating enhanced crystallographic orientation (41). To elucidate the interaction mechanism between [Bcmim]Cl and the perovskite precursor components, we performed ¹H nuclear magnetic resonance spectroscopy in dimethyl sulfoxide-*d*₆ (DMSO-*d*₆) (fig. S12). This chemical shift upon introduction of PbI₂ implies coordination between the acidic proton of [Bcmim]⁺ and Pb²⁺, likely via halide-bridged ion pairs or Pb-N interactions. Such coordination suggests that [Bcmim]Cl alters the local chemical environment of Pb²⁺, forming intermediate complexes that regulate the crystallization pathway, consistent with our SEM and GIWAXS analyses. To elucidate the impact of [Bcmim]Cl on trap-mediated nonradiative recombination, we performed space-charge-limited current (SCLC) measurements on electron-only and hole-only devices (fig. S13). In electron-only devices, incorporation of [Bcmim]Cl led to a pronounced reduction in trap-filled limit voltage (V_{TFL}) from 0.21 to 0.19 V, corresponding to a decrease in electron trap density from 1.18×10^{15} to 1.07×10^{15} cm⁻³. A similar trend was observed in hole-only devices, with the trap density decreasing from 3.72×10^{15} cm⁻³ ($V_{TFL} = 0.66$ V) to 2.93×10^{15} cm⁻³ ($V_{TFL} = 0.52$ V) upon additive treatment (Fig. 2C). These results confirm that [Bcmim]Cl effectively passivates trap states in the bulk, thereby suppressing trap-assisted recombination pathways (42).

In LEO, solar cells undergo rapid thermal cycling more than 15 times per day due to alternating sunlight and eclipse periods, with device temperatures potentially varying from about +90° to -90°C during each orbit (43–46). Such fluctuations impose strict stability requirements, underscoring the need for thermally robust perovskite photovoltaics for space applications. To evaluate the optoelectronic stability of the WBG perovskite films under representative thermal and operational stresses, we performed in situ photoluminescence

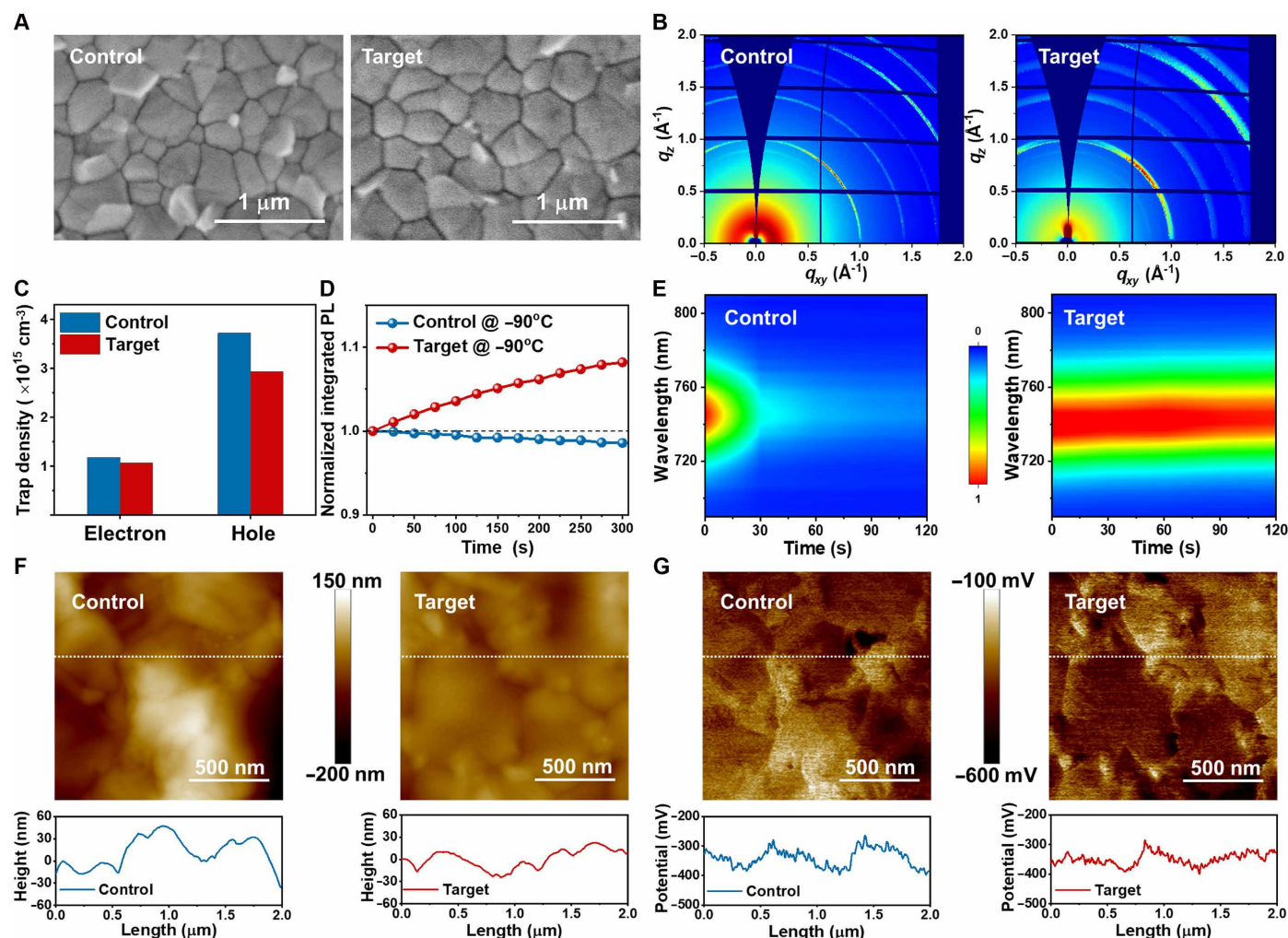


Fig. 2. Characterization of 1.72-eV WBG perovskite films with and without [Bcmim]Cl additive. (A) Top-view SEM images. (B) GIWAXS patterns. (C) Trap density extracted from SCLC measurements in electron- and hole-only devices. (D) Integrated normalized PL intensity overtime at -90°C . (E) In situ PL spectra under continuous 1-sun illumination at RT. (F) Atomic force microscopy (AFM) images ($2\ \mu\text{m}$ by $2\ \mu\text{m}$), together with the corresponding line profiles. (G) KPFM images ($2\ \mu\text{m}$ by $2\ \mu\text{m}$), together with the corresponding line profiles.

(PL) measurements under three conditions: continuous monitoring at -90°C , continuous 1-sun illumination, and combined illumination and heating at 90°C . In the -90°C test, the normalized PL intensity of the target film gradually increased, whereas that of the control slightly decreased (Fig. 2D and fig. S14), suggesting lower nonradiative loss in the treated film. As shown in Fig. 2E, the target film exhibited slower PL quenching under 1-sun illumination than the control, indicative of inhibition of halide phase segregation and defect formation. Under the combined light and heat at 90°C condition (fig. S15), the treated films retained substantially higher PL intensity than the control, confirming that the additive provides synergistic stabilization against photothermal degradation and enhances phase stability (47, 48). The [Bcmim]Cl-treated films further exhibited a markedly smoother morphology (Fig. 2F), with the surface roughness decreasing from 55.7 to 20.5 nm in R_q . Together with the SEM observation of enlarged grains and a narrower grain-size distribution, these results indicate that [Bcmim]Cl promotes more uniform crystallization and suppresses local morphological and

electronic heterogeneity in the WBG perovskite film (49). In addition, the average surface potential shifts from -349 to -375 mV through the Kelvin probe force microscopy (KPFM) test (Fig. 2G), reflecting an increase in the perovskite film's work function. This upward shift of the Fermi level enhances energy-level alignment with the electron transport layer (C_{60}), facilitating more efficient electron extraction at the interface (50).

Performance and stability of engineered WBG perovskite devices

We used a single-junction p-i-n device architecture consisting of indium tin oxide (ITO)/self-assembled monolayer (SAM)/perovskite/ C_{60} /bathocuproine (BCP)/Ag and first examined the effects of [Bcmim]Cl additive loading on device performance. The corresponding performance metrics are summarized in fig. S16. As demonstrated, the device with 1.0 mg/ml exhibited the best performance. The introduction of the additive increases V_{OC} from 1.281 to 1.292 V and improves the fill factor (FF) from 84.09 to 84.77%, boosting the

champion PCE from 23.23 to 24.00% (Fig. 3A), demonstrating a highly competitive performance for 1.72 eV-based PSCs (Fig. 3B and table S2). This performance was independently verified by an accredited institute (Wuxi Institute of Inspection, Testing, and Certification), with a certified steady-state power output (SPO) efficiency of 23.58% (fig. S17). The target device exhibits a markedly reduced hysteresis index (HI) of 0.5%, compared with 4.6% for the control device. The corresponding EQE spectrum matches well with the enhanced J_{SC} (fig. S18). The histogram of efficiencies in Fig. 3C shows that the target devices exhibit higher average efficiency and a narrower distribution than the control. Figure S19 reveals that the performance

enhancement in target devices primarily stems from improvements in V_{OC} and FF, attributed to smoother film morphology, mitigated trap formation, and suppressed nonradiative recombination. To elucidate the underlying cause of the FF enhancement in the target device, we further examined its loss mechanisms. Figure 3D substantiates that the FF gain of the target device is primarily enabled by reduced nonradiative recombination, which aligns well with the additive-induced improvements in film quality and defect passivation. To further probe the recombination behavior, we performed light intensity-dependent V_{OC} measurements (Fig. 3E). The ideality factor decreased from $1.38\text{ kT}/q$ in the control to $1.18\text{ kT}/q$ in the target device, indicative of

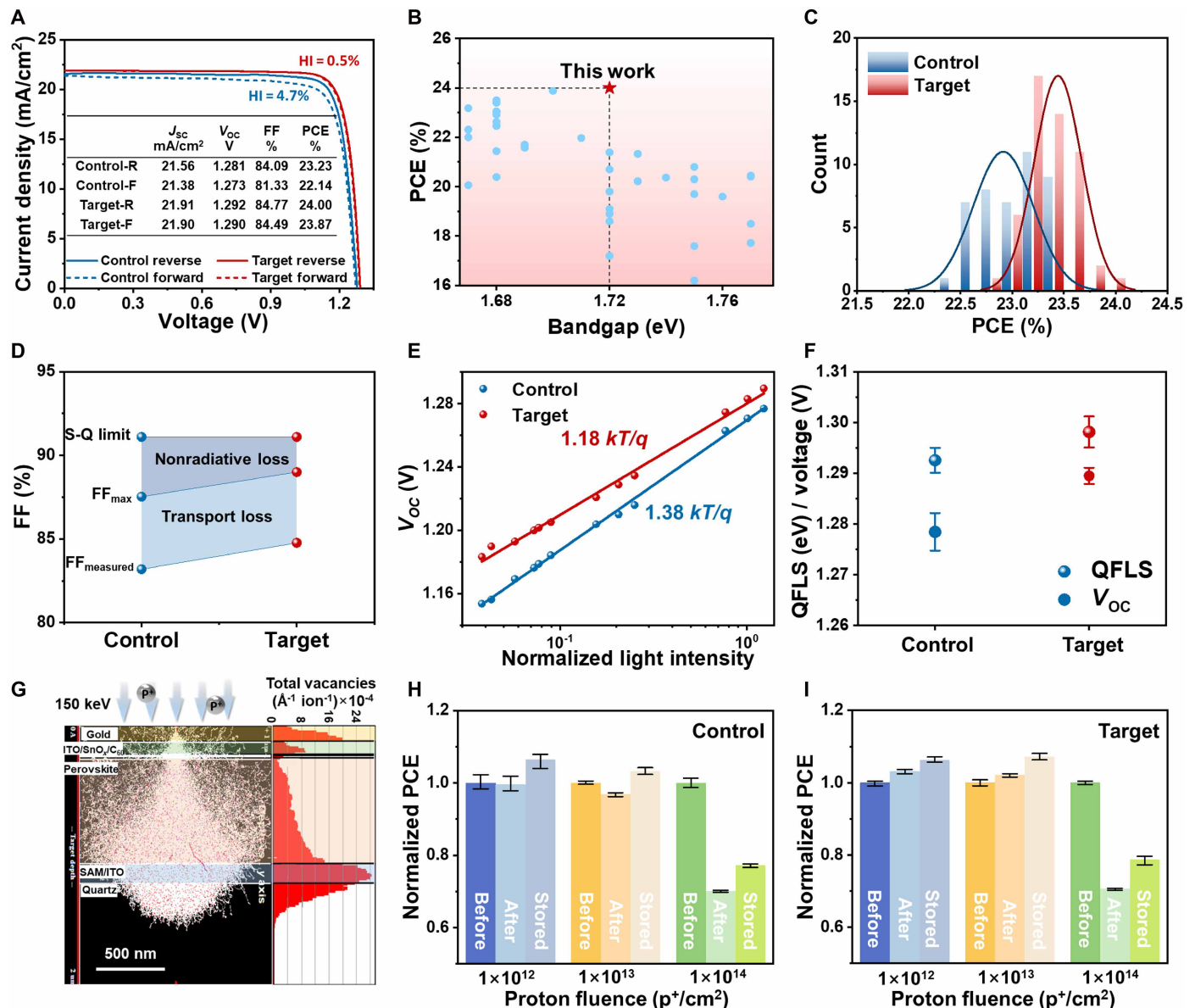


Fig. 3. Photovoltaic performance and stability enhancement via [Bcmim]Cl additive. (A) J - V characteristics of control and target ([Bcmim]Cl-treated) devices measured under AM1.5G illumination. (B) Summary of literature-reported PCEs for 1.67- to 1.77-eV WBG PSCs. (C) PCE histogram of control and target PSCs (D). Calculated FF loss of control and target PSCs. (E) Light intensity-dependent V_{OC} for both control and target PSCs. (F) QFLS of the control and target samples with the structure: glass/ITO/SAM/perovskite, and measured V_{OC} of the corresponding devices. (G) SRIM Monte Carlo simulation of 150-keV proton trajectories incident on the perovskite device stack. (H and I) Normalized PCE evolution of the control (H) and target (I) single-junction PSCs under 150-keV proton irradiation at fluences of 1×10^{12} , 1×10^{13} , and 1×10^{14} p⁺/cm², measured before irradiation, after irradiation, and after 5 days of dark storage.

reduced trap-assisted recombination. The target films exhibited a quasi-Fermi level splitting (QFLS) of $1.298 \text{ eV} \pm 0.3\%$, slightly higher than the $1.292 \text{ eV} \pm 0.2\%$ observed in the control (Fig. 3F), consistent with the V_{OC} enhancement in the devices. These results confirm that [Bcmim]Cl simultaneously improves both film morphology and reduces trap density, leading to reduced nonradiative recombination and enhanced charge extraction.

To assess the intrinsic operational stability of the 1.72-eV WBG PSCs, we used the 90°C light-soaking maximum power point tracking (MPPT) test as a simplified, accelerated protocol to emulate the combined stress environment relevant to space operation. As illustrated in fig. S20, the control device degraded rapidly to 80%, whereas the target device exhibited negligible performance loss, maintaining $\sim 100\%$ of its initial efficiency over the same period. These results highlight the enhanced thermal and photostability imparted by [Bcmim]Cl. This trend is also consistent with the PL measurements. Furthermore, [Bcmim]Cl was found to enhance the stability of the perovskite precursor solution. As evidenced by fig. S21, no visible color change or precipitation was observed in the solution after aging for one week, indicating suppressed degradation in the solution phase. Devices fabricated from these aged solutions exhibited minor performance losses (fig. S22), with PCE decreasing slightly from $23.57 \pm 0.23\%$ to $23.38 \pm 0.17\%$. This improved shelf-life of the precursor solution offers greater flexibility for large-scale manufacturing and enhances batch-to-batch consistency in device fabrication.

To further evaluate the cold-end stability of the WBG perovskite devices, we conducted temperature-dependent photovoltaic measurements while cooling from room temperature (RT) to -90°C . The devices were mounted on a temperature-controlled stage within a sealed chamber, where cooling was achieved by circulating liquid nitrogen. All measurements were performed under vacuum to eliminate moisture effects and prevent condensation. The PCE of the target perovskite device (figs. S23 and S24) increased $\sim 5\%$ at -90°C , whereas the control device exhibited an $\sim 9\%$ decrease relative to its initial PCE. The most pronounced difference lies in FF, which drops substantially in the control device at low temperature but remains much more stable in the target device. This behavior, together with the time-dependent PL enhancement observed only for the target film at -90°C , suggests that the target device not only undergoes less nonradiative loss in the absorber but also maintains more efficient charge extraction under cold-end operation (51).

To clarify the intrinsic radiation response of the perovskite absorber, we performed 150-keV proton irradiation experiments. Low-energy protons remain a stringent probe of displacement damage-dominated radiation effects, although they are expected to be strongly attenuated by cover-glass shielding under practical encapsulation conditions (20). Stopping and Range of Ions in Matter (SRIM) simulations (Fig. 3G) were then performed to determine the penetration depth and damage profile of 150-keV protons in the device stack. The simulations showed that 150-keV protons traverse the perovskite layer to a depth of about $1.12 \mu\text{m}$, with the highest energy loss and total-vacancy generation concentrated near the perovskite/SAM/ITO buried interface. We therefore examined unencapsulated single-junction perovskite devices with a quartz/ITO/SAM/perovskite/ $\text{C}_{60}/\text{SnO}_2/\text{ITO}/\text{Au}$ structure, matching the tandem top-subcell architecture, under proton irradiation from the metal-electrode side. At fluences of 1×10^{12} and $1 \times 10^{13} \text{ p}^+/\text{cm}^2$, both control and target devices showed clear post-irradiation recovery after 5 days of dark storage, whereas the target devices exhibited almost no immediate degradation and

achieved higher retention factors than the control devices (1.07 and 1.08 versus 1.04 and 1.02, respectively; Fig. 3, H and I, and fig. S25). At $1 \times 10^{14} \text{ p}^+/\text{cm}^2$, recovery was incomplete for both devices, with retention factors of 0.78 and 0.80 for the control and target, respectively, indicating that radiation-triggered healing is fluence dependent and becomes limited under severe proton damage. The stronger losses in FF and J_{SC} at high fluence, together with the EQE suppression (fig. S26), are consistent with combined interfacial extraction losses and bulk collection losses induced by proton-generated defects. In contrast, the single-junction silicon devices showed no evidence of self-healing upon storage and underwent severe, essentially irreversible degradation, accompanied by a dramatic loss of photo-response in the EQE spectra after irradiation at all three fluences (figs. S27 and S28).

Monolithic WBG perovskite/p-Si tandem architectures

We further fabricated monolithic TSCs by integrating the WBG perovskite top cell with the p-Si bottom cell. A schematic stack of this TSC is shown in Fig. 4A. In our study, the bottom cell consisted of a silicon heterojunction device fabricated on a $100\text{-}\mu\text{m}$ -thick p-type crystalline silicon wafer, following the procedure detailed in the Supplementary Materials. This thickness was chosen to simultaneously reduce device mass and improve radiation tolerance, both of which are critical for space applications. Cross-sectional SEM image (Fig. 4A) reveals the layered structure of the tandem stack, with the perovskite layer exhibiting uniform coverage and conformal contact over the textured surface of the silicon bottom cell. Tandem devices were fabricated in both small-area (aperture area: 0.95 cm^2) and large-area (aperture area: 12.56 cm^2) formats (Fig. 4B). Under AM1.5G illumination, the small-area device achieved a champion PCE of 32.07%. Leveraging the same device architecture and processing conditions, we then scaled up to the large-area TSC, which maintained a high PCE of 30.58%. The minimal drop in efficiency upon scaling indicates excellent process uniformity and negligible area-related losses. This high consistency is attributed to the enhanced film homogeneity achieved through the additive-assisted perovskite crystallization strategy. To validate the reproducibility of TSCs, we fabricated and characterized over 50 devices (Fig. 4C), and the resulting PCEs exhibited high uniformity.

Next, to evaluate device performance under extraterrestrial illumination conditions, the 12.56-cm^2 -area tandem cell was characterized under the AM0 spectrum (1353 W/m^2) using a calibrated system at an accredited third-party testing facility (Tianjin CETC New Energy Research Institute Co., Ltd.). The corresponding test report is provided in fig. S29. Under AM0, the certified device maintained a high PCE of 27.49%, with a V_{OC} of 2.003 V, J_{SC} of 22.44 mA/cm^2 , and FF of 82.74% (Fig. 4D), representing a leading efficiency reported so far for perovskite/silicon TSCs under space-relevant conditions (Fig. 4E and table S3). The SPO of the device closely matched its performance (fig. S30). The corresponding EQE characteristic is shown in Fig. 4F. The measured J_{SC} aligned well with the integrated photocurrent values obtained from the EQE spectra (24.13 mA/cm^2 for the top cell and 24.07 mA/cm^2 for the bottom cell), with the slight reduction in the J_{SC} likely attributed to the shadowing effect of the front grid electrodes. Notably, the integrated photocurrents of the top and bottom subcells under AM0 were nearly identical, indicating excellent current matching across the full spectrum. This spectral balance is particularly crucial under AM0 conditions, where the photon flux in the short-wavelength region (300 to 700 nm) is substantially higher

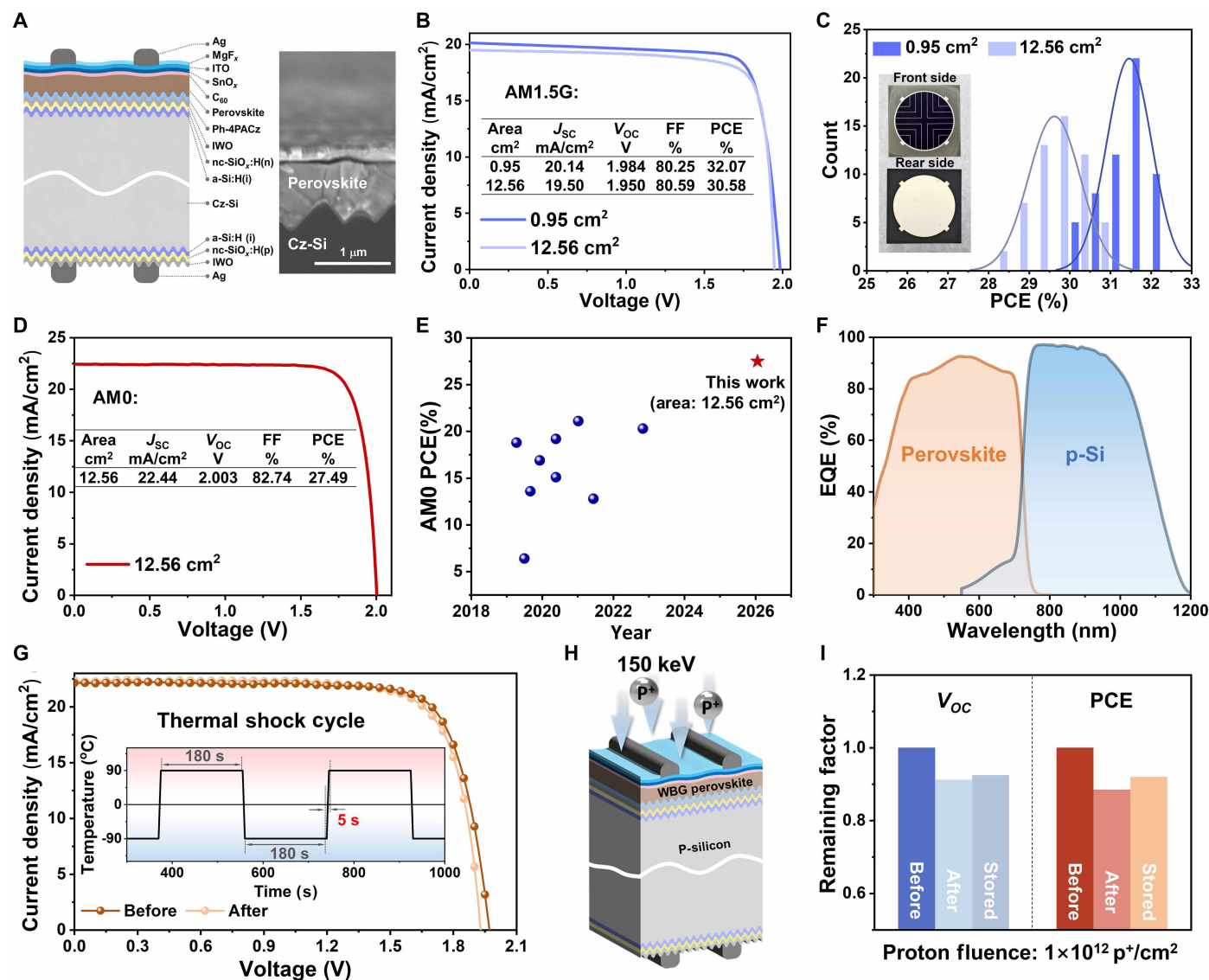


Fig. 4. Device performance of WBG perovskite/p-Si TSCs. (A) Schematic of the monolithic WBG perovskite/p-Si TSC and corresponding cross-sectional SEM image. (B) Representative J - V curves of TSCs under AM1.5G illumination. (C) PCE distribution from over 50 devices for each size category; inset shows a photograph of the 12.56-cm²-area device (the size of the Si substrate: 5 cm by 5 cm). (D) The certified J - V curve of the 12.56-cm² device measured under the AM0 spectrum (1353 W/m²). (E) Summary of reported AM0 PCEs for perovskite-based solar cells from the literature. (F) EQE spectra of champion devices with integrated J_{SC} values shown for AM1.5G and AM0 spectra. (G) J - V curves of the 12.56-cm²-area TSC under AM0 condition before and after six thermal-shock cycles (inset: thermal-shock profile for representative cycle segment). (H) Schematic illustration of 150-keV proton irradiation on the unencapsulated monolithic WBG perovskite/p-Si TSC. (I) Remaining factors of V_{OC} and PCE of the tandem device measured before irradiation, after proton irradiation at a fluence of $1 \times 10^{12} \text{ p}^+/\text{cm}^2$, and after 5 days of dark storage in the glovebox.

than under AM1.5G illumination. In contrast, under AM1.5G, the top and bottom cells generated 19.45 and 20.45 mA/cm², respectively, highlighting the increased challenge of current matching when optical design is not optimized for space spectra. As a result, the efficiency achieved in this work represents a highly competitive AM0 performance for WBG perovskite/p-Si TSCs.

To assess reliability under rapid thermal transients relevant to orbital environments, the encapsulated 12.56-cm² monolithic TSC was subjected to thermal-shock cycling between -90° and 90°C . The transition time between the two temperature extremes occurred within ~ 5 s, corresponding to an effective temperature ramp of about 36°C/s (inserted in Fig. 4G), thereby providing a highly accelerated

probe of transition-induced thermomechanical stress. This extreme range not only covers representative near-space thermal extremes, but also applies an accelerated stress to screen potential failure modes such as interfacial delamination, microcrack formation, and encapsulation failure induced by coefficient of thermal expansion mismatch. The J - V characteristics before and after six thermal-shock cycles are shown in Fig. 4G, with the corresponding test profile inset. After the thermal cycles, the J_{SC} remained virtually unchanged, while the V_{OC} and FF decreased slightly. As a result, the PCE remained at 96.8% of its initial PCE. Next, we examined the response of TSCs to low-energy proton irradiation. Under 150-keV proton irradiation at a fluence of $1 \times 10^{12} \text{ p}^+/\text{cm}^2$, the monolithic TSCs without encapsulation

(Fig. 4H) retained 93% of their initial PCE, indicating only moderate degradation at this low-energy proton fluence, with the performance loss arising mainly from a decrease in V_{OC} (remaining factor ~ 0.90 ; Fig. 4I and fig. S31). In addition, under continuous 1-sun white light-emitting diode lighting at $45 \pm 5^\circ\text{C}$ and $70 \pm 10\%$ relative humidity, the encapsulated tandem device retained $\sim 95\%$ of its initial PCE after 550 hours (fig. S32). However, the control device underwent a much faster degradation, retaining only $\sim 60\%$ of its initial efficiency after the same aging duration.

Near-space in situ performance benchmarking

To evaluate the near-space operational performance of the TSCs, we carried out a system-level validation campaign using a high-altitude balloon platform. This approach exposes devices to near-space stressors (Fig. 5A), including low pressure, intense ultraviolet irradiation (AM0), rapid thermal transients, and platform vibration, which are difficult to reproduce simultaneously in terrestrial simulation facilities. A 12.56-cm^2 encapsulated WBG perovskite/p-Si TSC

was integrated into a dedicated radiosonde balloon payload. The encapsulation was designed to support stable operation during ascent and near-space exposure. Specifically, a $100\text{-}\mu\text{m}$ radiation-resistant cover glass was bonded to the front side using a space-grade silicone encapsulation layer, providing mechanical protection during handling and flight while serving as the primary protective barrier against the harsh near-space environment. In addition, a 200-nm SiO_x layer was deposited by e-beam evaporation between the silicone encapsulation layer and the top electrode. This inorganic interlayer serves as a barrier against moisture and oxygen ingress, further improving the environmental robustness of the encapsulation structure. The payload was subsequently carried into near space for in situ I-V acquisition under authentic AM0 solar irradiance and low-pressure boundary conditions. The overall balloon architecture and the integrated flight platform are shown in Fig. 5 (B and C).

The flight profile (launch at 07:00, apogee $\sim 29.7\text{ km}$ at 09:35, and recovery at $\sim 10:30$) and the corresponding device temperature evolution (-29.32° to 17.11°C) are shown in figs. S33 and S34. The

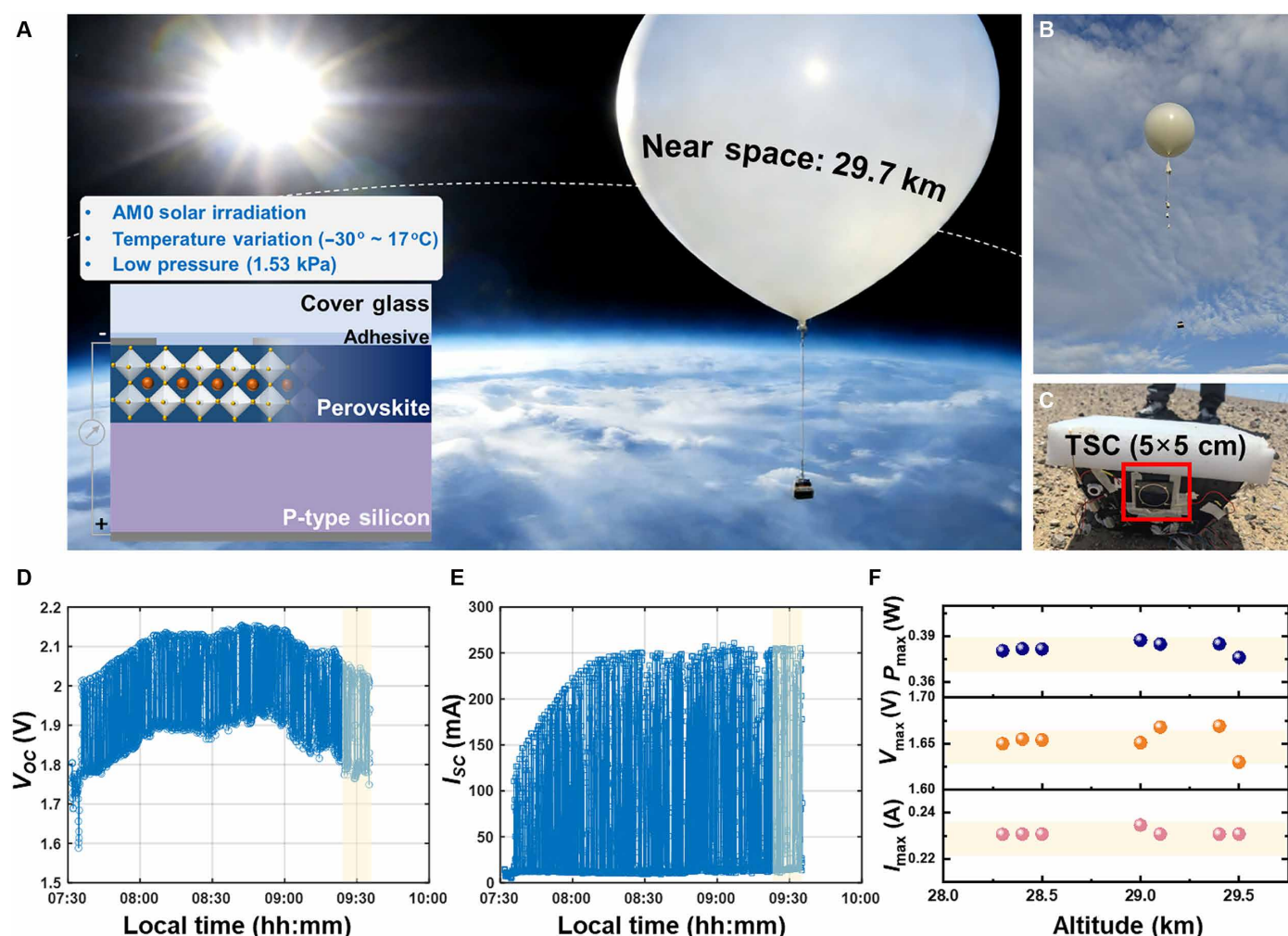


Fig. 5. Near-space high-altitude balloon platform for in situ monitoring of WBG Perovskite/p-Si TSCs. (A) Schematic illustration of the high-altitude balloon system operating in near space. Insert: Device schematic and representative environmental conditions during the near-space balloon flight. (B) Photograph of the balloon prior to launch. (C) Photograph of the platform integrating the TSC and the in-flight measurement setup. (D) Real-time V_{OC} of the TSC recorded during the flight. (E) Real-time I_{sc} of the TSC recorded during the flight. (F) Summary of the MPP parameters (I_{max} , V_{max} , and P_{max}) extracted from the high-altitude segments (28.3 to 29.7 km) during the flight.

corresponding evolution of V_{OC} and I_{SC} is presented in Fig. 5 (D and E), respectively. Notably, after the balloon entered the near-space altitude corridor (20 to 30 km), both V_{OC} and I_{SC} exhibited only minor fluctuations around steady mean values. To quantitatively assess operational stability under representative near-space conditions, we analyzed seven high-altitude data segments (S1-S7, corresponding to altitudes of 28.3 to 29.7 km), marked by the shaded regions in the flight profile. These windows were intentionally selected from the quasi-steady float phase, where ambient pressure (1.29 to 1.53 kPa) and device temperature (5.52 to 16.14°C) were well constrained, thereby minimizing confounding transients associated with rapid ascent/descent. The corresponding full J - V curves acquired under real AM0 solar illumination, together with the associated environmental conditions, are provided in fig. S35. Across this defined near-space window, the encapsulated TSC exhibited highly reproducible electrical output and a stable maximum power output, with P_{max} reaching 387.4 mW at the best-performing segment (Fig. 5F). The observed minor fluctuations in V_{OC} (~2.05 V) and I_{SC} (~250 mA) align with expected variations in solar incidence angle and atmospheric transmission during flight. No monotonic decay or distortions in the I - V characteristics were observed, indicating that both encapsulation and electrical interconnections remained intact throughout the near-space exposure. The high-resolution, traceable I - V curves were acquired under authentic low-pressure and AM0-irradiance conditions. These data establish a critical benchmark for calibrating ground-based simulators with flight operation and for guiding future platform-level evaluation of space-deployable photovoltaics.

DISCUSSION

We demonstrated a WBG perovskite/p-Si tandem configuration evaluated under AM0 illumination. A 1.72-eV WBG perovskite top cell was compositionally tuned and integrated with a p-type crystalline silicon bottom cell, enabling improved current matching under the AM0 spectrum and enhanced operational stability under high-energy electron and proton exposure. Incorporation of a multifunctional ionic-liquid additive, [Bcmim]Cl, during perovskite film formation improved crystallinity and suppressed nonradiative losses. It also mitigated phase segregation and enhanced thermal and photostability, enabling a 1.72-eV single-junction perovskite with a PCE of 24.0% (certified 23.58%). The resulting tandem devices achieved PCEs of 32.07% (0.95 cm²) under AM1.5G and a certified 27.49% (12.56 cm²) under AM0 conditions. Electron irradiation experiments with 1-MeV electrons demonstrated that devices based on the 1.72-eV top cell maintained higher post-irradiation performance compared to those with lower-bandgap absorbers. The observed stability enhancement was associated with improved current continuity between subcells and reduced mismatch following radiation exposure, indicating that appropriate bandgap selection in the perovskite absorber can mitigate radiation-induced performance loss in tandem configurations. Beyond ground-based qualification, we further provided near-space in situ evidence via a high-altitude balloon campaign, where the encapsulated tandem delivered stable electrical output at 20- to 30-km altitude with a maximum power of 387.4 mW and reproducible J - V characteristics across multiple high-altitude segments, supporting preserved functional integrity of the encapsulation and interconnect stack during flight exposure. This work establishes a practical design framework that addresses three key challenges in the development of space-grade tandem photovoltaics,

including spectral mismatch under AM0, environmental instability of WBG materials, and radiation-induced degradation. The combination of bandgap-tuned perovskite design, additive-assisted film stabilization, and integration with radiation-hardened silicon provides a viable route to scalable, durable photovoltaic systems for orbital platforms, satellite arrays, and planetary surface missions.

MATERIALS AND METHODS

Materials

Anhydrous DMSO (≥99.9%), dimethylformamide (DMF; 99.8%), anisole (99.7%), and isopropanol (99.5%) were purchased from Sigma-Aldrich. Cesium iodide (CsI; 99.99%), lead(II) iodide (PbI₂; 99.99%), lead(II) bromide (PbBr₂; 99.99%), lead(II) chloride (PbCl₂; 99.99%), (4-(3,6-diphenyl-9H-carbazol-9-yl) butyl)phosphonic acid (Ph-4PACz; >98%), piperazinium iodide (PI), C₆₀ (>99.5% purity), and BCP (>99.5% purity) were purchased from Xi'an Polymer Light Technology Corp. Methylammonium chloride (99.9%) and formamidinium iodide (FAI) were purchased from Youxuan Tech. ITO glass was purchased from South China Xiangcheng Corp. The ceramic ITO target (In₂O₃/SnO₂ 90/10 wt %) was purchased from China General Research Institute for Nonferrous Metals. The chemicals and solvents were used without further purification.

Device fabrication

Preparation of PSCs

Glass/ITO substrates are cleaned by ultrasonication in deionized water, acetone, and isopropyl alcohol (IPA) for 15 min each. For the hole transport layer (HTL), Ph-4PACz solution in ethanol (0.5 mg/ml) is spin-coated in a nitrogen glovebox at 3000 rpm for 30 s and annealed at 100°C for 10 min. The 1.5 M perovskite precursor solution of Cs_{0.22}FA_{0.78}Pb(I_{0.85}Br_{0.15})₃ (for 1.68 eV) and Cs_{0.22}FA_{0.78}Pb(I_{0.8}Br_{0.2})₃ (for 1.72 eV) was prepared by dissolving a mixture of CsI, FAI, PbBr₂, and PbI₂ in a mixed solution of DMF and DMSO (v/v 4:1). Additional 5% MAPbCl₃ and 3% PbCl₂ were added. Before use, the precursor solution was stirred at 50°C for 2 hours until clarified and then filtered through a 0.22-μm polytetrafluoroethylene membrane. Then, the perovskite films were spin-coated in a nitrogen glovebox at 4000 rpm for 45 s, with 5-s acceleration and 300 μl of anisole as the antisolvent introduced at ~25 s into the spin-coating process. The perovskite layer is then annealed at 100°C for 15 min. A 1 mg/ml solution of PI in IPA is spin-coated at 5000 rpm for 30 s and annealed for 10 min at 100°C. Then, substrates were transferred to a C₆₀-BCP deposition chamber. Last, C₆₀ (20 nm) and BCP (6 nm) were deposited under high vacuum (<1.0 × 10⁻⁶ torr) at a deposition rate of 0.2 Å/s. After deposition of the C₆₀/BCP layer, 100-nm-thick Ag was thermally evaporated under high vacuum (<1.0 × 10⁻⁶ torr). The effective area of the devices was 6.723 mm².

Preparation of p-Si solar cells

The fabrication process began by etching crystalline silicon wafers in a potassium hydroxide solution to create a pyramidal surface texture, followed by standard Radio Corporation of America (RCA) cleaning. The wafers were then immersed in a 3 to 5% hydrofluoric acid solution for 1 min to remove the native oxide. After wet-chemical cleaning, an intrinsic hydrogenated amorphous silicon (i-a-Si:H) layer was deposited on the front side of the textured wafer, followed by an n-type hydrogenated nanocrystalline silicon oxide (n-nc-SiO:H) layer using very high-frequency plasma-enhanced chemical vapor deposition with a gas mixture of PH₃/CO₂/SiH₄ to establish the built-in

electric field. On the rear side, i-a-Si:H and p-type hydrogenated nanocrystalline silicon (p-nc-Si:H) layers were sequentially deposited using a B_2H_6/SiH_4 gas mixture to form the back surface field structure. Transparent conductive oxide layers, ~80 nm thick, consisting of WO_3 -doped In_2O_3 (IWO), were deposited on both sides by reactive plasma deposition. Last, metal grid electrodes were fabricated via screen-printing. All cell fabrication steps were performed at temperatures around 200°C.

Preparation of monolithic perovskite/p-Si TSCs

Both sides of the p-Si were textured to enhance absorption in the near-infrared region, while the front surface featured a fine microscale texture with an average height of ~800 nm to promote uniform deposition of the spin-coated perovskite layer. A 10-nm IWO layer served dual purposes as the recombination interface and as a foundation for the subsequent SAM. The device architecture comprised p-Si/IWO/Ph-4PACz/perovskite/ $C_{60}/SnO_x/ITO/Ag/MgF_x$ layers. The Ph-4PACz solution dissolved in ethanol (0.5 mg/ml) was spin-coated on the cleaned Si substrate at 3000 rpm for 30s, followed by annealing at 100°C for 5 min as HTL. The thickness of the transparent IWO layer on Si was decreased to ~10 nm to improve the transparency. The perovskite top cell was deposited onto the underlying silicon bottom cell using the same procedure as for the single-junction perovskite devices. The spin-coating processes were conducted in an N_2 -filled glovebox. The devices were then completed by thermally evaporating 20 nm of C_{60} , followed by depositing 9 nm of SnO_x via atomic layer deposition. An ~80-nm ITO layer was sputtered at 60 W, and 1 μm of Ag fingers and busbars were thermally evaporated on top of the ITO. Last, a 100-nm MgF_x layer was deposited on the samples by thermal evaporation.

Characterization

J-V characteristics were measured using a Keithley 2400 source meter under simulated illumination. For the AM1.5G condition, the measurements were conducted on a class AAA solar simulator (Enlitech). The light intensity was calibrated prior to each measurement with a National Institute of Metrology (China) calibrated KG5-filtered Si reference cell. The AM0 solar simulator consists of two light sources, namely, a xenon lamp and a halogen lamp, which together provide highly accurate AM0 irradiance. The AM0 simulator is spectrally adjustable and uses the EQE along with spectral mismatch factors to determine the irradiance. The triple-junction GaAs module was measured with a photocurrent ratio of 1.0 between each junction pair, with less than 1% error in the simulator spectrum for each junction. The *J-V* curves of the devices are measured under the AM1.5G spectrum (at 25°C and 1000 W/m² irradiance) and the AM0 spectral irradiance (at 25°C and 1353 W/m²). Current-voltage scans were performed in the forward and reverse directions at 50 mV/s. In addition, we tracked the maximum power point (MPP) using home-made feedback software.

Electron irradiation measurement

One mega-electron volt electron irradiation was carried out at RT, using an electron accelerator (model: DD1.2, manufactured by Shanghai Pioneer) from the Institute of Technical Physics of Heilongjiang Academy of Sciences, situated in Harbin, China. All electron irradiation devices were unencapsulated and performed from the top metal-electrode side. The fluences of 1-MeV electron beam were ranged from 2×10^{13} to 2×10^{14} e⁻/cm², with the flux of 1×10^{11} e⁻/cm².s.

Proton irradiation measurement

Proton irradiation (150 keV) was carried out under a vacuum of ~10⁻⁴ Pa using a proton accelerator at the Laboratory for Space Environment and Physical Sciences, Harbin Institute of Technology, China. The irradiation fluences of the proton beam were 1×10^{12} , 1×10^{13} , and 1×10^{14} p⁺/cm², with a constant proton flux of 1×10^{10} p⁺/cm².s. The proton-irradiated single-junction perovskite devices had a quartz/ITO/SAM/perovskite/ $C_{60}/SnO_x/ITO/Au$ structure, which was adopted to preserve consistency with the top subcell architecture in the monolithic tandem device. All devices were unencapsulated, and proton irradiation was performed from the metal-electrode side.

Suns-*V*_{OC} measurements

The incident light intensity was varied by modulating the simulator intensity, and the stabilized *V*_{OC} was measured. In all cases, the temperature was set to 25°C using a temperature-controlled stage. The ideality factor (*n*) was extracted from the slope of the *V*_{OC} versus $\ln(\text{light intensity}, I)$ plot, according to the equation $n = \frac{q}{kT} \cdot \frac{dV_{OC}}{d\ln I}$, where *q* is the elementary charge, *k* is the Boltzmann constant, *T* is the absolute temperature, and *I* is the incident light intensity. This parameter provides insight into the dominant recombination mechanism within the device.

GIWAXS measurements

GIWAXS measurements were carried out at beamline 1W1A of Beijing Synchrotron Radiation Facility with an incident beam energy at 8 keV (the fixed grazing angle is 0.2°). The beam center and sample-to-detector distance were calibrated with LaB6. The samples were directly prepared on clean Si substrates under the same conditions as the device preparation.

QFLS measurements

QFLS measurements were performed on samples in air using a Lumin QY manufactured by Shanghai Ideaoptics Co., Ltd. and Shenzhen Yiyang Technology Co., Ltd. Photoluminescence quantum efficiency (PLQE) data were obtained using a 0.013-mW laser operating at an excitation wavelength of 520 nm.

PL measurements

The in situ PL spectra were acquired using a custom-built system developed by the Yangtze Delta Institute of Optoelectronics, Peking University. PL imaging was performed under ambient conditions using a LabRAM HR Evolution Raman spectrometer (Horiba Jobin Yvon) equipped with a 50× fluorescence objective lens (Nikon) and an excitation wavelength of 530 nm. During the measurements, the samples were mounted on a temperature-controlled stage, and the sample temperature was precisely monitored using a temperature sensor.

KPFM measurements

KPFM measurements were performed using a Bruker Dimension Icon atomic force microscope integrated with a custom-built KPFM setup.

SCLC measurements

Trap density (*n*_t) was quantitatively determined from SCLC measurements, using *V*_{TFL}. According to the SCLC model, *n*_t is calculated using the equation: $n_t = \frac{2\epsilon\epsilon_0 V_{TFL}}{qL^2}$, where ϵ is the dielectric constant of the

perovskite, ϵ_0 is the vacuum permittivity, V_{TFL} is the trap-filling limit voltage, q is the elementary charge, and L is the film thickness.

Simulation

Optical simulation

TSCs are simulated using the transfer matrix method. The refractive indices for each material (n, k) comprising the different layers of the TSC analyzed in this study are taken from literature or extracted by ellipsometry fitting.

Electron irradiation simulation

Monte Carlo simulations were performed using the CASINO software (version 2.51). The multilayer structure was modeled based on the actual device stack, comprising 13 distinct layers, including MgF_x (100 nm), ITO (80 nm), C_{60} (20 nm), SnO_x (9 nm), perovskite (800 nm), SAM (Ph-4PACz, 5 nm), IWO (10 nm), amorphous and nanocrystalline silicon-based buffer layers (i-a-Si:H, n-nc-SiO:H, and p-nc-Si:H), and a crystalline silicon (Cz-Si, 100 μm) base, followed by a rear IWO contact (80 nm). Each layer was defined with its corresponding thickness, density, and elemental composition based on fabrication parameters. The simulation was configured with an incident electron energy of 1 MeV (1000 keV), a beam radius of 5 nm, and 100,000 incident trajectories to ensure statistical robustness. Backscattering and energy deposition processes were enabled, while secondary electron and x-ray processes were excluded. Displacement damage calculations were activated to estimate non-ionizing energy loss contributions. The spatial resolution for energy deposition was set to 0.1 μm to obtain a fine-grained depth-resolved profile.

Proton irradiation simulation

TRIM simulations using the SRIM package were performed to evaluate the penetration depth and collision-induced damage from low-energy proton irradiation in the multilayer perovskite device stack. The simulations were carried out in the full-damage-cascades mode with 150-keV protons incident from the Au side at a normal incidence angle (0°). The device architecture used in the simulation was modeled as Au (120 nm)/ITO (80 nm)/ SnO_x (9 nm)/ C_{60} (20 nm)/perovskite (800 nm)/SAM (5 nm)/ITO (150 nm)/quartz (2 mm), where the quartz substrate was approximated as SiO_2 , and the NiO_x layer was omitted for simplification. The densities used in the simulation were 19.31 g/cm^3 for Au, 7.14 g/cm^3 for ITO, 4.50 g/cm^3 for SnO_x , 1.65 g/cm^3 for C_{60} , 4.34 g/cm^3 for the perovskite absorber, 1.40 g/cm^3 for SAM, and 2.20 g/cm^3 for quartz. The perovskite layer was described using a composition of $\text{Cs}_{0.22}\text{FA}_{0.78}\text{Pb}(\text{I}_{0.8}\text{Br}_{0.2})_3$, corresponding to an atomic composition of Cs (2.10 at %), C (7.46 at %), H (37.28 at %), N (14.91 at %), Pb (9.56 at %), I (22.94 at %), and Br (5.74 at %). Unless otherwise noted, the simulations were used to extract the projected range, longitudinal straggling, and depth-dependent collision/vacancy distributions of the incident protons within the full device stack.

High-altitude balloon experiment and in situ electrical acquisition

The balloon-flight system consisted of a radiosonde balloon, a cutter, a parachute, a solar-cell calibration unit, a radiosonde, an iridium communication terminal, and rope-based tether lines. The cutter was used to separate the balloon from the parachute either at the end of the mission or under abnormal flight conditions, enabling controlled termination of the flight. The parachute served as part of the mechanical connection chain during ascent and was deployed after

balloon separation for payload recovery. Its size was selected according to the recovery mass and the target landing speed, with the criterion that the landing speed remained below 7 m/s. To meet the payload-mass limit of the radiosonde balloon, the solar-cell calibration unit was designed with a lightweight tetrahedral architecture, with a sun sensor mounted on each face to monitor incident solar irradiance during flight. The main circuit board of the high-precision solar cell test system was housed within the unit. For in-flight testing, an encapsulated TSC was integrated into the payload using a radiation-resistant laminate structure of cover glass/space-grade silicone/ SiO_x /cell/space-grade silicone/glass. Onboard electrical characterization was performed using an external-bias excitation scheme based on a programmable MOSFET load. During each scan, the MOSFET gate voltage was stepped through a predefined sequence to sweep the operating point of the tested solar cell across the full I-V range. At each bias point, the cell terminal voltage and the voltage drop across a precision shunt resistor were synchronously recorded, and the device current was reconstructed from the shunt measurement. Each scan consisted of 100 bias points, yielding a full 100-point I-V curve, and all data were stored locally on an onboard SD card. To ensure measurement accuracy under flight conditions, the circuit used precision sampling resistors, low-noise signal conditioning, and 16-bit analog-to-digital conversion, giving voltage and current resolutions of 0.1 mV and 0.01 mA, respectively. The flight campaign was conducted within a 07:00 local launch window to improve irradiance stability and reduce meteorological perturbations affecting payload attitude, thermal balance, and illumination geometry. The payload reached an apogee of ~ 29.7 km at 09:35, after which a programmed cut-down sequence separated the calibration unit from the balloon, followed by parachute-assisted descent and recovery.

Supplementary Materials

The PDF file includes:

Figs. S1 to S35
Tables S1 to S3
Legend for data S1

Other Supplementary Material for this manuscript includes the following:

Data S1

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